TECHNICAL REPORT

LUN-TR-20261030-2352D6

# LUNARIS

LUNAR UNCHARTED NAVIGATION, AUTONOMY & RESOURCE INTELLIGENCE SYSTEM

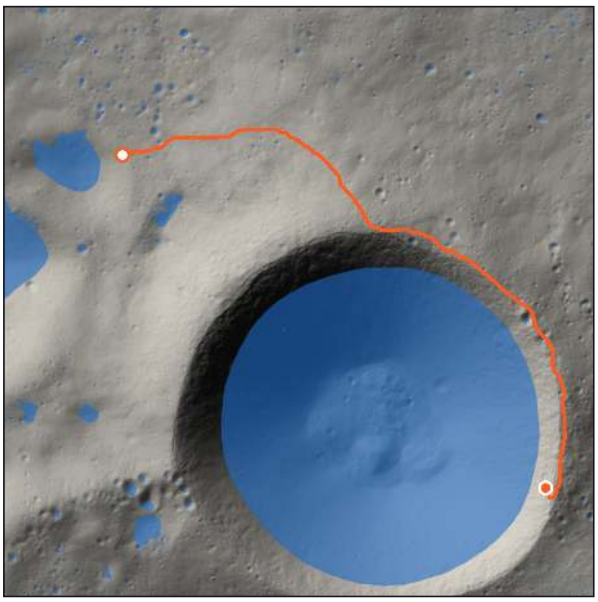
# Traverse and Volatile-Prospect Assessment

## Shackleton · Lunar South Polar Region

Issued 30 October 2026 · Prepared with the LUNARIS analysis pipeline

### DOCUMENT CONTROL

| | |
|---|---|
| Report number | LUN-TR-20261030-2352D6 |
| Date of issue | 2026-10-30 09:35 UTC |
| Study region | Shackleton |
| Dataset | Shackleton crater & south pole — LOLA 20 m/px (real data) |
| Region centre | 89.67 S, 129.78 E · D = 20.9 km |
| Analysis grid | 800 x 800 cells @ 40 m |
| Configuration | 0cc97e48aa17f087 |
| Dataset checksum | 5de0c23e315fcde4 |
| Primary sources | NASA LRO: LOLA (PGDA), Diviner (PDS), LROC NAC; LOLA illumination models |
| Distribution | Unrestricted — derived from NASA open data |



LOLA shaded relief, PSR (blue) and candidate traverse (orange)



Lunar surface data: NASA/GSFC/ASU/UCLA/MIT

EXECUTIVE SUMMARY

## Summary of findings

This report documents an orbital-data assessment of the Shackleton study area using the LUNARIS pipeline. Terrain derived from LOLA laser altimetry classifies 32% of the analysis area as low mobility hazard and 25% as impassable at the adopted slope and roughness thresholds. Permanently shadowed regions occupy 25.7% of the window. A hazard-weighted candidate traverse of 39.4 km was identified, with a maximum slope of 24.8° and no impassable cells crossed. The volatile investigation-priority model flags 23.7% of the area as candidates for further investigation. A traverse simulation of the NASA VIPER (Volatiles Investigating Polar Exploration Rover) returned **MARGINAL**.

| ROUTE LENGTH | MAX SLOPE ON ROUTE | PERMANENT SHADOW | ROVER SIMULATION |
|---|---|---|---|
| 39.4 km | 24.8° | 25.7% | MARGINAL |

SECTION 01

## Introduction and scope

LUNARIS is a decision-support system for permanently shadowed regions (PSRs) near the lunar poles. It combines laser altimetry, thermal-model products, illumination models and orbital imagery into traceable data layers, plans rover traverses over a documented cost surface, and evaluates them with a terramechanics-based rover simulation. Every layer in this report is classified as measured, derived, model estimate or illustrative, and cells without valid data are treated as unknown rather than safe.

**Study region.** 32 x 32 km window in south polar stereographic coordinates (x -12..20 km, y -16..16 km) containing Shackleton crater, the south pole, and the Shackleton–de Gerlache connecting ridge. Named features are listed only because this window actually covers them. Named features inside the window (IAU gazetteer): Shackleton crater (rim clipped ~0.9 km at the window's lower edge), Lunar south pole, Shackleton–de Gerlache ridge (partial).

SECTION 02

## Data sources

**Table 1.** Input data products and checksums.

| Layer | Product | Instrument | Units | SHA-256 |
|---|---|---|---|---|
| elevation | LDEM_80S_20MPP_ADJ | LRO LOLA | m, height relative to 1737.4 km reference sphere | e627615dfe12 |
| elevation uncertainty | LDEM_80S_20MPP_ADJ_ERR | LRO LOLA | m | 8d5eae88b9a6 |
| effective resolution | LDEM_80S_20MPP_ADJ_EFFRES | LRO LOLA | m | d629a3694c55 |
| slope reference | LDSM_80S_20MPP_ADJ | LRO LOLA | deg | 4947f038f1b8 |
| psr mask | LPSR_80S_20MPP_ADJ | LRO LOLA (illumination modelling on DEM) | 1 = PSR, 0 = not PSR | 9fcdb10874ae |
| image | M139729686LE_20MPP | LRO LROC NAC | radiance factor (I/F), as distributed | 298e3f0169f0 |
| illumination avg | AVGVISIB_85S_060M_201608 | LRO LOLA (illumination modelling) | fraction of time the solar disc is visible (DN x 0.00004) | 2776c3ce120c |
| earth visibility avg | AVGVISIB_85S_060M_201608_EARTH | LRO LOLA (visibility modelling) | fraction of time the Earth disc is visible (DN x 0.00004) | 75e7d11d52ed |
| thermal max | DLRE_PRP_SOUTH.TAB:temp_max | LRO Diviner (DLRE) | K | a11bbc10c16e |
| thermal avg | DLRE_PRP_SOUTH.TAB:temp_avg | LRO Diviner (DLRE) | K | 6c0ad441478e |
| ice stability depth | DLRE_PRP_SOUTH.TAB:ice_depth | LRO Diviner (DLRE) | m (depth where sublimation < 1 kg m^-2 Gyr^-1; deeper than 2.87 m is encoded as 999) | f76d09b21c1b |

**Table 2.** Layers not ingested for this study region.

| Layer | Status |
|---|---|
| hydrogen | not provided by this dataset |

**Citation.** Barker et al. (2023) PSJ 4:183 doi:10.3847/PSJ/acf3e1; Paige et al. (2010) Science 330:479

SECTION 03

## Methods

Rasters are ingested with checksum verification, aligned to the elevation grid and block-averaged to the analysis resolution (cells require at least 75% valid sub-cells). Slope uses the Horn (1981) operator; roughness is the RMS residual from a local least-squares plane. Hazard classes take the worse of the slope and roughness classes; producer uncertainty raises a cell by one class. Traverses use an 8-connected A* search with an admissible octile heuristic over a per-metre cost multiplier.

**Table 3.** Processing record.

| Stage | Detail |
|---|---|
| Ingest | elevation — LDEM_80S_20MPP_ADJ.TIF |
| Ingest | elevation uncertainty — LDEM_80S_20MPP_ADJ_ERR.TIF |
| Ingest | effective resolution — LDEM_80S_20MPP_ADJ_EFFRES.TIF |
| Ingest | slope reference — LDSM_80S_20MPP_ADJ.TIF |
| Ingest | psr mask — LPSR_80S_20MPP_ADJ.TIF |
| Ingest | image — M139729686LE_20MPP.TIF |
| Ingest | illumination avg — sun_avg_visibility.tif |
| Ingest | earth visibility avg — earth_avg_visibility.tif |
| Ingest | thermal max — prp_temp_max.tif |
| Ingest | thermal avg — prp_temp_avg.tif |
| Ingest | ice stability depth — prp_ice_depth.tif |
| Align | elevation uncertainty — identical grid (no resampling) |
| Align | effective resolution — identical grid (no resampling) |
| Align | slope reference — identical grid (no resampling) |
| Align | psr mask — identical grid (no resampling) |
| Align | image — resampled (nearest) onto elevation grid |
| Align | illumination avg — resampled (bilinear) onto elevation grid |
| Align | earth visibility avg — resampled (bilinear) onto elevation grid |
| Align | thermal max — identical grid (no resampling) |
| Align | thermal avg — identical grid (no resampling) |
| Align | ice stability depth — identical grid (no resampling) |
| Resample | factor 2 — nan-aware block mean |
| Analyse | 1.2 s · configuration 0cc97e48aa17f087 |

SECTION 04

# Terrain assessment

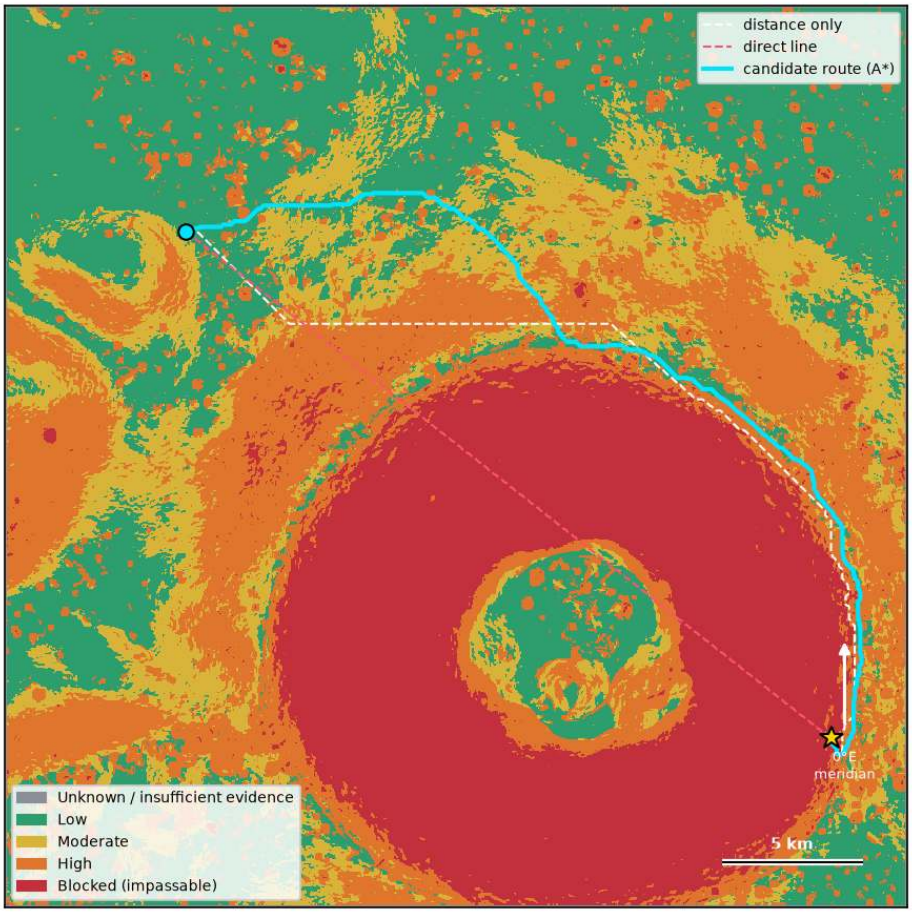



**Figure 1.** Mobility hazard classes with the candidate traverse (cyan), distance-only alternative and direct line. Up is the 0°E meridian direction of the polar stereographic grid.

**Table 4.** Hazard class distribution over the analysis area.

| Hazard class | Area fraction |
|---|---|
| Unknown / insufficient evidence | 1.0% |
| Low | 31.8% |
| Moderate | 16.6% |
| High | 25.3% |
| Blocked (impassable) | 25.3% |

Derived slope agrees with the producer slope map with r = 0.998 and a median absolute difference of 0.36° (a consistency check on the same source elevation model).

**Boulder-scale hazards.** grid is 40 m/px; rover-scale boulders (<10 m) are sub-pixel. Requires native-resolution NAC/ShadowCam imagery (~0.5–2 m/px) and a labelled training/evaluation set.

SECTION 05

# Traverse planning

**Table 5.** Candidate traverse compared with rejected alternatives.

| Metric | Candidate (hazard-weighted) | Distance-only | Direct line |
|---|---|---|---|
| Length (m) | 39,434 | 36,984 | 30,336 |
| Path cost | 102,420 | 193,096 | impassable |
| Maximum slope (°) | 24.8 | 24.9 | 34.1 |
| Mean slope (°) | 7.2 | 14.4 | 21.6 |
| Cells in PSR | 1 | 20 | 330 |
| Blocked cells crossed | 0 | 0 | 282 |
| Total ascent (m) | 2,844 | 3,141 | 5,906 |

Path cost = sum over moves of step length x mean per-metre cost multiplier of the two cells; a value equal to the length means flat, certain, low-hazard terrain all the way. Relative units — not energy, time or risk probability.

Start 89.6789° S, 325.04° E; goal 89.3407° S, 120.08° E (planetocentric).

**Table 6.** Hazard exposure along the candidate traverse.

| Hazard class along route | Cells |
|---|---|
| Unknown / insufficient evidence | 0 |
| Low | 535 |
| Moderate | 114 |
| High | 202 |
| Blocked (impassable) | 0 |

SECTION 06

# Volatile investigation priority

The priority model combines psr, thermal_max, ice_stability_depth with explicit weights; layers not used: hydrogen. A cell is assessed only when thermal evidence is present. Assessed fraction: 100.0%.

**Table 7.** Investigation-priority class distribution.

| Priority class | Area fraction |
|---|---|
| Not assessed (insufficient evidence) | 0.0% |
| Lower relative priority | 73.4% |
| Moderate relative priority | 2.9% |
| Candidate for further investigation | 23.7% |

Under 9 weight-perturbation scenarios, 31.5% of assessed cells change class at least once; the ranking should be read with this sensitivity in mind.

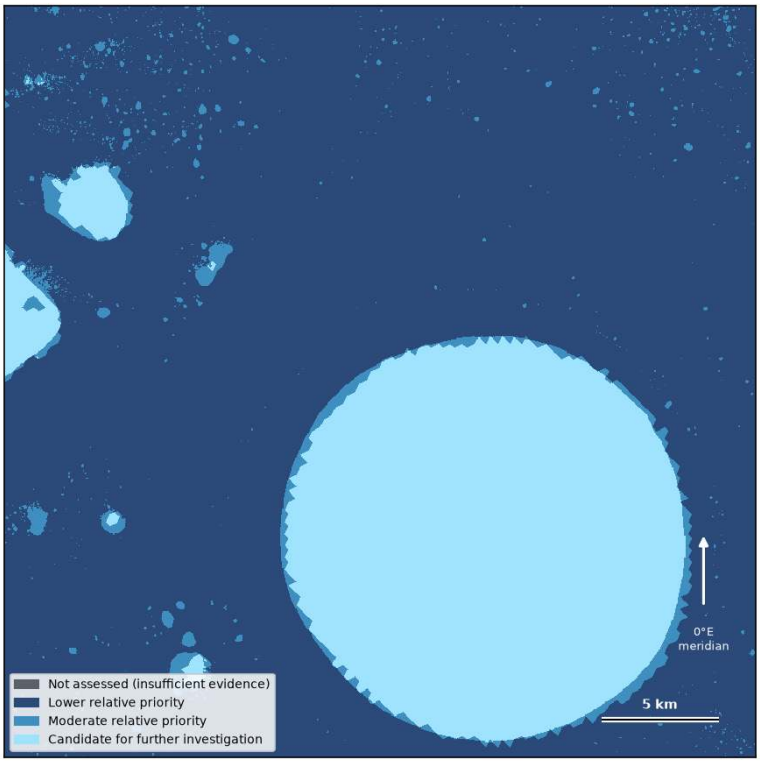



**Figure 2.** Relative investigation priority for water ice (ranking of evidence, not a detection).

SECTION 07

## Rover traverse simulation

The NASA VIPER (Volatiles Investigating Polar Exploration Rover) was driven along the candidate traverse over the native-resolution elevation model on Lunar regolith — Apollo LRV design values, with Bekker–Wong wheel–soil forces, static stability and shadow-time checks. Verdict: **MARGINAL**.

**Table 8.** Rover simulation results.

| Quantity | Value |
|---|---|
| Distance driven / route length | 39.43 / 39.43 km |
| Drive time (continuous) | 128.8 h |
| Maximum slope / pitch / roll | 27.8° / 26.0° / 21.1° |
| Wheel slip, maximum / mean | 74% / 14% |
| Maximum sinkage | 25.3 mm |
| Minimum static stability margin | 7.7° |
| Time in PSR (longest continuous) | 0.23 h (0.17 h) |
| Shadow energy / usable battery | 141 / 4878 Wh |
| Loose-regolith sensitivity case | FAIL — immobilised: slope 16.3° needs more thrust than the soil can provide at 100 % slip (9.68 km) |

**Table 9.** Simulation events.

| Event | Distance (km) |
|---|---|
| slope 15.9° exceeds the 15° design requirement (1616 m of route) | 3.57 |
| slip 43% > 40% operational threshold (308 m of route) | 9.68 |
| static stability margin 9.4° < 10° | 38.71 |

Unpublished rover parameters adopted as assumptions: ground clearance m, cg height m, drive efficiency, battery reserve frac. Terrain is the LOLA DEM at its native 20 m grid (bilinear). Rocks, boulders and craters smaller than ~20-40 m are invisible to this simulation; VIPER's published 15 cm obstacle limit cannot be checked from this data. Wheel-soil forces use the Bekker-Wong model with Apollo LRV soil parameters (Costes et al. 1972). Real PSR regolith properties are unmeasured and may differ; see the loose-soil sensitivity run. Quasi-static model: no dynamics, wheel-level suspension articulation, slip-sinkage or turn-in-place time.

SECTION 08

## Assumptions

- Slope classes [10.0, 15.0, 25.0] deg and blocked >= 25.0 deg are prototype assumptions.
- Roughness high >= 2.0 m, blocked >= 6.0 m RMS (prototype).
- Cost weights {'slope': 8.0, 'roughness': 3.0, 'uncertainty': 4.0, 'psr': 1.5} are prototype choices; cost is relative, not energy or time.
- Ice-priority weights {'psr': 0.35, 'thermal_max': 0.4, 'ice_stability_depth': 0.25} with sensitivity +/-50%.
- Ground distance equals grid distance (polar stereographic scale error < 0.04 % within 2 deg of the pole).
- No-data and unknown cells are treated as blocked for routing.

SECTION 09

## Limitations

- Analysis grid 40.0 m/px: rover-scale obstacles (boulders, small craters) are not resolved.
- PSR map, temperatures and ice-stability depth are model products, each with its own epoch and resolution.
- No LEND hydrogen layer was used; no direct measurement of water ice is included.
- Illumination, communications line-of-sight and thermal survival are not modelled in the route.
- Layer 'LEND hydrogen' unavailable: not provided by this dataset

APPENDIX A

# Layer catalogue

**Table 10.** Analysis layers, epistemic status and method.

| Layer | Status | Units | Method |
|---|---|---|---|
| Elevation (LOLA) | measured | m | Block-mean 2x2 of source DEM (cells need >=75% valid sub-pixels). |
| Hillshade (visual aid) | illustrative | unitless | Lambertian shading of Horn gradients. |
| Slope | derived | degrees | Horn (1981) finite differences on the analysis grid; NaN propagates from no-data. |
| Roughness (detrended RMS) | derived | m | Least-squares plane removal per window (separable correlation). |
| DEM uncertainty (PGDA) | derived | m | Block mean. |
| DEM effective resolution | derived | m | Block mean. |
| Elevation flagged uncertain | derived | boolean | Threshold on producer uncertainty layers (prototype thresholds). |
| Slope — PGDA reference | derived | degrees | Block mean of 20 m slope. |
| Permanently shadowed regions | derived | fraction of cell | Block mean of the producer's 0/1 PSR map (illumination modelling on LOLA topography). |
| Annual max temperature (Diviner PRP) | derived | K | Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged. |
| Annual mean temperature, 2 cm depth (Diviner PRP) | derived | K | Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged. |
| Ice stability depth (Diviner PRP) | derived | m | Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged. |
| Average sun visibility (18.6-yr) | derived | fraction of time | LOLA horizon modelling, hourly over 18.6 yr; 60 m product resampled bilinearly, then block mean. |
| Average Earth visibility (18.6-yr) | derived | fraction of time | LOLA horizon modelling, hourly over 18.6 yr; 60 m product resampled bilinearly, then block mean. |
| LEND hydrogen | unavailable | - | - |
| Mobility hazard class | derived | class | Slope edges [10.0, 15.0, 25.0] deg, blocked >= 25.0 deg; roughness high >= 2.0 m, blocked >= 6.0 m. |
| Traversal cost multiplier | derived | cost per metre (>= 1) | See core/navigation/cost.py |
| Ice investigation priority (score) | model_estimate | 0–1 relative | core/ice_priority/evidence.py |
| Ice investigation priority (class) | model_estimate | class | Thresholds at 1/3 and 2/3. |
| Evidence coverage | derived | fraction of weight | - |
| Priority sensitivity (std) | derived | score std | One-at-a-time perturbation. |
| Image (LROC NAC) | measured | radiance factor (I/F), as distributed | Block mean for display. |

Prepared with LUNARIS (Lunar Uncharted Navigation, Autonomy & Resource Intelligence System). Configuration 0cc97e48aa17f087; dataset checksum 5de0c23e315fcde4; generated 2026-10-30T09:35:26 UTC.